



IVIM using convolutional neural networks predicts microvascular invasion in HCC

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Abstract

Objectives The study aimed to investigate the diagnostic performance of intravoxel incoherent motion (IVIM) diffusion-weighted magnetic resonance imaging for prediction of microvascular invasion (MVI) in hepatocellular carcinoma (HCC) using convolutional neural networks (CNNs).

Methods This retrospective study included 114 patients with pathologically confirmed HCC from December 2014 to August 2021. All patients underwent MRI examination including IVIM sequence with 9 *b*-values preoperatively. First, 9 *b*-value images were superimposed in the channel dimension, and a *b*-value volume with a shape of $32 \times 32 \times 9$ dimension was obtained. Secondly, an image resampling method was performed for data augmentation to generate more samples for training. Finally, deep features to predict MVI in HCC were directly derived from a *b*-value volume based on the CNN. Moreover, a deep learning model based on parameter maps and a fusion model combined with deep features of IVIM, clinical characteristics, and IVIM parameters were also constructed. Receiver operating characteristic (ROC) curve analysis was performed to assess the diagnostic performance for MVI prediction in HCC.

Results Deep features directly extracted from IVIM-DWI (0.810 (range 0.760, 0.829)) using CNN yielded better performance for prediction of MVI than those from IVIM parameter maps (0.590 (range 0.555, 0.643)). Furthermore, the performance of the fusion model combined with deep features of IVIM-DWI, clinical features (α -fetoprotein (AFP) level and tumor size), and apparent diffusion coefficient (ADC) (0.829 (range 0.776, 0.848)) was slightly improved.

Conclusions Deep learning with CNN based on IVIM-DWI can be conducive to preoperative prediction of MVI in patients with HCC.

Key Points

- Deep learning assessment of IVIM data for prediction of MVI in HCC can overcome the unstable and low performance of IVIM parameters.
- Deep learning model based on IVIM performs better than parameter values, clinical features, and deep learning model based on parameter maps.
- The fusion model combined with deep features of IVIM, clinical characteristics, and ADC yields better performance for prediction of MVI than the model only based on IVIM.

Keywords Hepatocellular carcinoma · Neoplasm invasion · Artificial intelligence · Diffusion magnetic resonance imaging

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Abbreviations

ADC	Apparent diffusion coefficient
CNN	Convolutional neural network
HCC	Hepatocellular carcinoma
IVIM	Intravoxel incoherent motion
MVI	Microvascular invasion

Introduction

Hepatocellular carcinoma (HCC) is one of the most common malignancies and is ranked as the fourth leading cause of cancer-related deaths worldwide [1]. Microvascular invasion (MVI), presence of micrometastatic HCC emboli within the vessels of the liver [2], is considered a critical determinant of recurrence after surgical resection and liver transplant [3–5]. Therefore, an accurate preoperative assessment of MVI is of great clinical importance in selecting appropriate surgical interventions.

Medical images including CT and MR images have been proposed to predict MVI in HCC based on morphological features [6, 7]; however, qualitative analysis is limited by differences in subjective interpretation. As a type of MRI that evaluates kinetic properties of tissues, diffusion-weighted imaging (DWI) is a promising method to predict MVI precisely [8, 9]. The major advantages are no need of using contrast agents and no involvement of ionizing radiation. Multi-*b*-value DWI with intravoxel incoherent motion (IVIM) can distinguish both true molecular diffusion and microcapillary perfusion using a biexponential model, which may be more accurate in determining tissue perfusion and diffusion than using apparent diffusion coefficient (ADC) calculated by monoexponential model. At present, the accuracy of IVIM-DWI in clinical application is mainly based on the quantitative characterization of IVIM parameters [10, 11]. However, IVIM parameters derived from the biexponential model are very sensitive to motion and artifacts in DWI [12], which significantly reduce the performance of IVIM-DWI in tumor characterization.

The development of machine learning and artificial intelligence has added a new scope to CT and MR imaging analyses [13]. One previous study elucidated the relationship between multiple *b*-values in DWI using a deep convolutional network, and the deep learning of feature representation learned from multiple *b*-value images was used for characterizing cancer lesions [14]. In addition, recently published research extracted the deep features of multiple *b*-value images in DWI using deep convolutional networks, which could better predict MVI in HCC [15]. To the best of our knowledge, a previous study has been reported proposing an attention-based deep learning model for processing IVIM data to improve the performance in tumor characterization, but without considering any clinical information and statistical analysis [16]. Hence, from a clinical point of view, we hope to independently assess the performance of deep learning models and clinical features

for MVI prediction based on IVIM-DWI, and further combine valuable clinical information with deep learning models to better characterize lesion in clinical practice.

To this end, the purpose of this study was to develop and evaluate a deep learning model based on IVIM-DWI to accurately predict preoperative MVI in HCC. Specifically, we planned to assess the fusion model based on IVIM-DWI, clinical characteristics, and IVIM parameters for better prediction of MVI.

Materials and methods

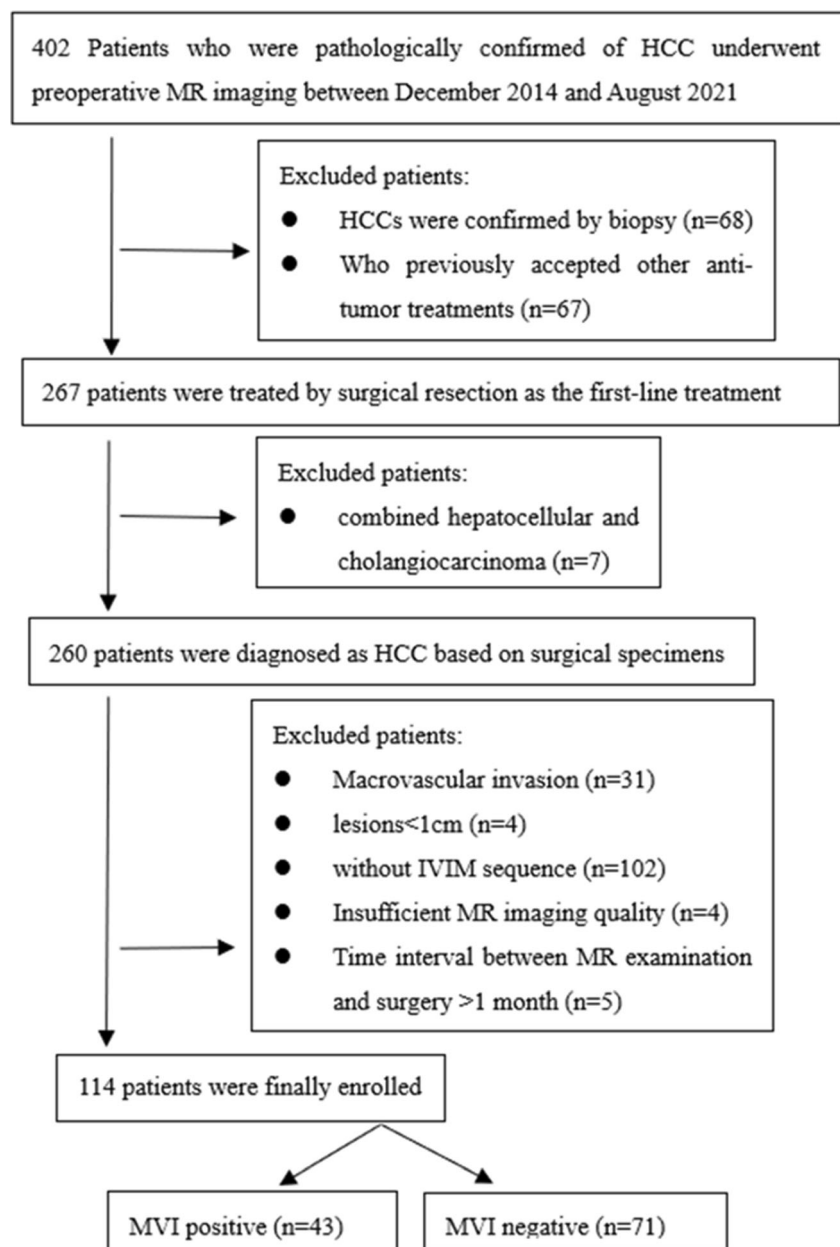
Patients

This retrospective study was approved by our institutional review board and the requirement for informed consent was waived. This study was strictly carried out in accordance with the Declaration of Helsinki. Patients who underwent MRI examination including IVIM-DWI sequence at our institution before surgery between December 2014 and August 2021 were consecutively included in this study. The inclusion criteria were as follows: (1) patients who underwent preoperative IVIM with 9 *b*-values at our institution within 1 month before surgery; (2) patients without macrovascular invasion determined by preoperative MRI; and (3) patients who underwent curative hepatectomy and were pathologically confirmed to have primary HCC. Exclusion criteria were as follows: (1) patients who previously received other antitumor treatments, such as transcatheter arterial chemoembolization or radiofrequency ablation; (2) patients who were suspected to have HCCs using biopsy; (3) patients whose MR imaging quality was insufficient for analysis; (4) patients whose MR images and clinical or pathological information were incomplete; and (5) patients with small HCC lesions of less than 1 cm diameter. Finally, 114 patients were included in the study; seven patients had two HCC lesions, and the remaining 107 patients had one HCC lesion each. And for cases with two HCC lesions, we only included the larger one in the analysis. Patients were randomly divided into a training cohort (*n* = 74; 66 males and 8 females) and a validation cohort (*n* = 40; 37 males and 3 females). A flow chart of patients' recruitment is described in Fig. 1.

Demographic and clinicopathological variables

Demographic and laboratory characteristics, including age, sex, hepatitis B status (positive or negative), α -fetoprotein (AFP) level, aspartate transaminase (AST), alanine transaminase (ALT), bilirubin, albumin, and platelet count, were retrieved from the electronic medical record system. Pathological information about tumor size, Edmondson-Steiner grade, MVI status, surgical resection margin, and cirrhosis of the noncancerous liver parenchyma were obtained from pathological reports. MVI was defined as the presence of tumor emboli in the

Fig. 1 Flow chart of patients' recruitment. HCC, hepatocellular carcinoma; IVIM, intravoxel incoherent motion; MVI, microvascular invasion



vascular space lined with endothelium, which was visible microscopically. Two dedicated pathologists with experience in abdominal pathology reviewed all specimen sections independently in a blinded manner (without knowing the patients' clinical data). Surgical specimen of each patient was obtained within 1 month of MRI examination. Of the 114 lesions, 71 had no pathologically confirmed MVI, and 43 had confirmed MVI. Demographic and clinicopathological variables of the training cohort and validation cohort are listed in Table 1.

MRI protocol

MRI examinations were performed using a 3.0-T MRI system (Achieva, Philips Health care) with a 16-channel body matrix

coil. The IVIM-DWI sequence (TR/TE = 1979 ms/57 ms, 132×114 matrix, 5-mm slice thickness, FOV = $400 \text{ mm} \times 352 \text{ mm} \times 176 \text{ mm}$, and slice gap = 0.5 mm) with 9 b -values ($b = 0, 10, 20, 40, 80, 200, 400, 600$, and 1000 s/mm^2) was performed using a respiratory trigger with a single-shot echo planar sequence. The 3.0-T MRI was equipped with an RF dual transmission source system to guarantee the reliability of IVIM.

IVIM imaging analysis

All IVIM-DWI images were transferred to a workstation (Diffusion Weighted Imaging Kit, GE Healthcare) for postprocessing. The ADC map was generated from the monoexponential model using the equation $S_b/S_0 = \exp(-b$

Table 1 Clinical characteristics and IVM parameter values in the training and validation cohorts

Characteristic	Training cohort (<i>n</i> = 74)		Validation cohort (<i>n</i> = 40)		<i>P</i> _{inter}
	MVI(+)	MVI(−)	MVI(+)	MVI(−)	<i>P</i> _{intra}
Age	50.61 ± 11.72	51.52 ± 10.30	44.07 ± 10.34	50.08 ± 11.79	0.111
Sexual					> 0.999
Male	23 (82.1)	43 (93.5)	14(93.3)	23 (92.0)	0.744
Female	5 (17.8)	3 (6.5)	1(6.7)	2 (8.0)	
Hepatitis B					0.633
Positive	26 (92.9)	43 (93.5)	14(93.3)	21 (84.0)	0.317
Negative	2 (7.1)	3 (6.5)	1(6.7)	4 (16.0)	
AFP level					0.031
≤ 8.1 ng/ml	7 (25.0)	22 (47.8)	2(13.3)	15 (60.0)	0.597
> 8.1 and < 400 ng/ml	14 (50.0)	19 (41.3)	6(40.0)	5 (20.0)	
≥ 400 ng/ml	7 (25.0)	5 (10.9)	7(46.7)	5 (20.0)	
AST, U/L	26.5 (21.3, 33.8)	21.5 (18.8, 28.3)	21(19.27)	26 (20.28.5)	0.422
ALT, U/L	25.5 (19.3, 39.0)	24.5 (21.8, 38.5)	24(17.39)	24 (17.5.30)	0.525
TBIL, μmol/L	13.4 (8.1, 17.1)	12.9 (9.5, 16.3)	12.6(9.19.8)	12.7 (10.4, 18.7)	0.720
Albumin, g/L	40.90 ± 3.46	39.90 ± 3.12	40.51 ± 5.37	38.90 ± 3.22	0.244
Platelet count (10 ⁹ /L)	163.5 (108.8, 212.5)	160 (138.5, 201.3)	174(99.214)	152 (99.228)	0.783
Edmondson-Steiner grade					< 0.001
Edmondson I–II	5 (17.9)	24 (52.2)	2(13.3)	19 (76.0)	0.172
Edmondson III–IV	23 (82.1)	22 (47.8)	13(86.7)	6 (24.0)	
Tumor size, cm	2.6 (1.7, 4.6)	2.0 (1.5, 2.9)	2.9(2.2, 4.8)	2.3 (1.7, 3.1)	0.074
ADC (10 ^{−3} mm ² /s)	1.125 ± 0.266	1.236 ± 0.334	1.074 ± 0.190	1.344 ± 0.303	0.005
<i>D</i> (10 ^{−3} mm ² /s)	0.918 ± 0.364	1.062 ± 0.390	0.934 ± 0.226	1.069 ± 0.335	0.178
<i>D</i> * (10 ^{−3} mm ² /s)	53.7 ± 22.9	51.3 ± 21.7	45.2 ± 17.6	58.8 ± 30.5	0.126
<i>f</i>	0.247 ± 0.093	0.219 ± 0.130	0.172 ± 0.090	0.247 ± 0.129	0.053

Age, albumin, and parameter values are expressed as mean ± SD and other continuous variables are presented as median (interquartile range). Categorical variables are presented as number (percentage) *MVI* microvascular invasion, *AFP* α-fetoprotein, *AST* aspartate transaminase, *ALT* alanine transaminase, *TBIL* total bilirubin

$\times \text{ADC}$), where S_b and S_0 represented the diffusion-weighted signal intensity that were obtained with b -values of b and 0, respectively. The biexponential model from an IVIM-DWI sequence was expressed using the following equation according to Le Bihan et al [17]: $S_b/S_0 = (1 - f) \cdot \exp(-bD) + f \cdot \exp[-b(D + D^*)]$, where D was the true diffusion coefficient, which represented pure molecular diffusion; D^* was the pseudodiffusion coefficient representing perfusion-related diffusion; f was the perfusion fraction linked to microcirculation. The regions of interest (ROIs) were manually drawn by a radiologist with 2 years of experience in abdominal radiology on DWI images with $b1000$ using publicly available software ImageJ (imagej.nih.gov/ij, ImageJ 1.52), and then confirmed by a senior radiologist with 8 years of experience in abdominal radiology. The ROI of each patient was placed on a maximum representative slice in HCC. Then, the ADC, D , D^* , and f values were calculated using parametric maps with the outlined ROIs by ImageJ.

Image processing and data augmentation

The tumor area of each b -value image of each patient was manually drawn for further analysis. Due to the large slice thickness of IVIM-DWI, only the axial views of tumor areas were extracted as image samples. Specifically, the ROIs were manually drawn by a radiologist with 2 years of experience in abdominal radiology independently, and then confirmed by a

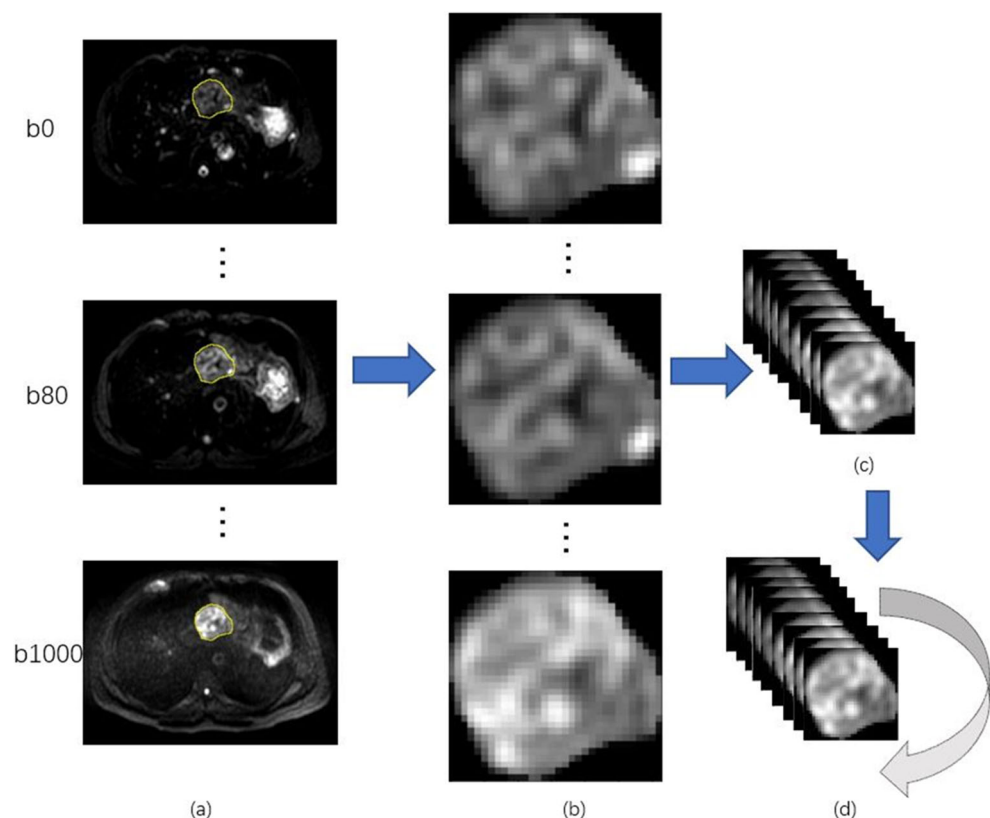
senior radiologist with 8 years of experience in abdominal radiology. Then, we extracted tumor areas from original images using MATLAB (The MathWorks, Inc.). Due to the various sizes of HCC lesions, all image samples were normalized to 32×32 . Then, tumor areas from 9 b -value images were superimposed in the channel dimension to obtain a b -value volume with a shape of $32 \times 32 \times 9$ dimension to use as the input.

The process of deep learning typically requires thousands of samples to train the network. Therefore, an image resampling method was performed for data augmentation to generate more samples from a limited tumor sample area for training the deep learning network. The data augmentation of tumor areas was conducted using the in-house software implemented by MATLAB. The training image samples were randomly rotated 50 times for data augmentation. The training image samples of the series of b -value images served as the input of the deep learning model, and MVI status obtained from the pathological report was considered the gold standard output for the deep learning model to predict. The process of image processing is shown in Figure 2.

Deep learning with a CNN and implementation

We used the LeNet-5 model to predict the MVI status in HCC based on IVIM-DWI. As shown in Figure 3, the input

Fig. 2 The process of image processing. **a** Original 9 b -value images with outlined ROIs, **(b)** the the extracted tumor areas from original images normalized to 32×32 , **(c)** a b -value volume with a shape of $32 \times 32 \times 9$ dimension obtained by superimposing 9 b -value images, and **(d)** data augmentation by random rotation of 50 times



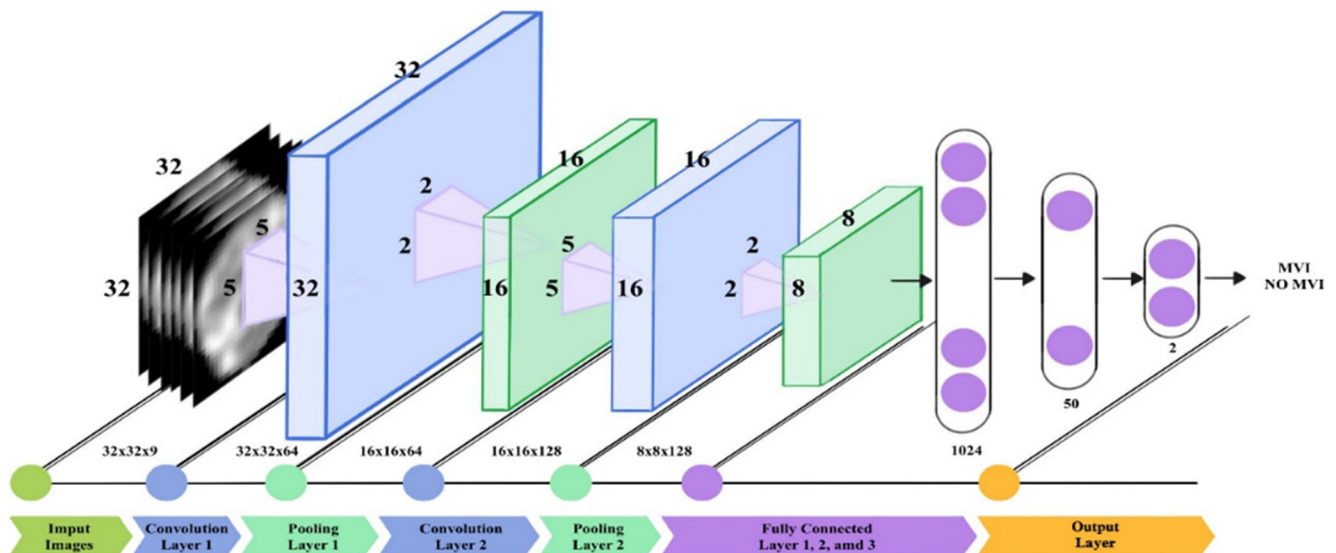


Fig. 3 The framework of the proposed deep learning network for MVI prediction in HCC

dimension was the $32 \times 32 \times 9$ IVIM-DWI sequence, which consisted of 9 *b*-value images from the axial view of the tumor region. Typically, there were 2 convolutional layers, 2 maximum pooling layers, and 3 fully connected layers in the deep learning model. The output used softmax for activation. First, the convolution feature map of the original image was obtained to determine the convolutional layer, by convolving the input image with a convolutional filter. Both convolutional layers had a 3×3 convolutional kernel. Each convolutional layer was followed by a max-pooling layer, which reduced the dimensionality of each feature map. Then, three fully connected layers with 1024, 50, and 2 neurons were sequentially connected. Finally, the probability of classification to discriminate the MVI status in HCC lesions was generated from the “softmax” function. Deep features that were derived from the last fully connected layer were chosen for MVI prediction. The loss function of the deep learning model adopted the cross entropy function.

In addition, we also evaluated the deep learning model based on IVIM parametric maps. The input images were normalized tumor areas extracted from parameter maps using the ROIs drawn on DWI images. Finally, significant clinical characteristics and parametric values were also integrated with deep features of IVIM data to develop the fusion model.

The proposed network structure was implemented using the platform “TensorFlow” (<https://tensorflow.google.cn/install>), and the server graphics card was RTX2080ti. The loss function of the deep learning model was minimized by the gradient descent algorithm. The number of training iterations was 40 k, and the batch size was 64. To reduce the risk of overfitting, the technique of early stop was adopted. The learning rate was initialized to $1e-3$, and the decay of the learning rate was 0.98.

Statistical analysis

Two independent sample *t*-test, Mann–Whitney *U* test, chi-square test, or Fisher exact test were used to compare clinical characteristics and IVIM parameter values between the MVI-positive and MVI-negative groups in the training set and validation set, as appropriate. To evaluate the stability of the deep learning network and reduce measurement errors, the training and testing process were repetitively performed ten times. Receiver operating characteristic curve (ROC) analysis was used to evaluate the performance for MVI prediction in the validation cohort. The Delong test was adopted to compare the diagnostic performance of ROCs between clinical characteristics, parameter values, and deep learning models. A two-sided *p* value of < 0.05 was considered statistically significant. All statistical analyses were performed using IBM SPSS statistical software (version 26) and MedCalc software (version 19.7.1).

Results

Training cohort and validation cohort

This study ultimately selected 114 lesions, 43 of which were pathologically confirmed as MVI-positive and 71 of which were pathologically confirmed as MVI-negative. The dataset was divided into a training cohort and validation cohort to test the performance of the deep learning models for MVI prediction. In the training cohort, there were 74 HCC lesions with 28 MVI-positive lesions and 46 MVI-negative lesions, while in the validation cohort, there were 40 HCC lesions with 15 MVI-positive lesions and 25 MVI-negative lesions.

Performance of clinical characteristics and parameter values

The clinical characteristics and IVIM parameter values of patients with HCC in the training and validation cohorts are listed in Table 1. No significant differences were found in clinical variables and parameter values between the training and validation cohorts. The Edmondson-Steiner grade was significantly related to MVI in both the training ($p = 0.003$) and validation cohorts ($p < 0.001$), reflecting that MVI was more commonly present in Edmondson-Steiner grades III–IV than in Edmondson-Steiner grades I–II. Furthermore, the AFP level was also significantly different between MVI-positive and MVI-negative patients in the training ($p = 0.031$) and validation cohorts ($p = 0.011$). Tumor size differed significantly in the training cohort but was not confirmed in the validation cohort. For IVIM parameter values, only the ADC showed statistically significant difference in the validation cohort.

The diagnostic performance of ADC and clinical characteristics, including AFP level and tumor size, in the validation cohort with cutoff values determined in the training cohort are shown in Table 2. The AUCs of AFP level, tumor size, and ADC were 0.733 (95% CI 0.574–0.892), 0.607 (95% CI 0.426–0.787), and 0.640 (95% CI 0.468–0.812), respectively.

Performance of deep learning model

The indicators of performance evaluation of the deep learning models, including AUC, accuracy, sensitivity, and specificity, were expressed as median (range, minimum, and maximum) of ten repeated measurements in the validation cohort. Table 3 shows the diagnostic performance of deep learning models for MVI prediction. Deep features directly extracted from IVIM-DWI (0.810 (range 0.760, 0.829)) yielded better performance for MVI prediction than those from IVIM parameter maps (0.590 (range 0.555, 0.643)). Furthermore, the performance of the fusion model combined with deep features of IVIM-DWI, clinical features (AFP level and tumor size), and ADC was slightly improved. The AUC, accuracy, sensitivity, and specificity of the fusion model were 0.829 (range, 0.776, 0.848), 0.775 (range, 0.700, 0.800), 0.666

(range, 0.600, 0.860), and 0.870 (range, 0.760, 0.920), respectively. Figure 4 shows the ROC curves of the different deep learning models including 9 b -value images, parameter maps, and the proposed fusion framework for MVI prediction.

In addition, we applied the gradient-weighted class activation mapping (Grad-CAM) technique to the trained CNN to overcome the lack of diaphaneity in CNN prediction and the problem of drawing simple conclusions in diagnosis. The Grad-CAM algorithm visualizes the feature extraction during the learning process of CNN as heatmaps [18]. Figure 5 shows the heatmaps generated by the Grad-CAM that identified the pixels on which the CNN model was fixating its attention for MVI prediction in HCC lesions. The network fixated most of its attention on the tumor marginal region.

Performance comparison

After using the Delong test to statistically compare the diagnostic performance between the deep learning model based on IVIM and AFP level, tumor size, and ADC in the validation cohort, it was found that the CNN model based on IVIM achieved better predictive performance for MVI than ADC ($p = 0.046$) in the validation cohort statistically, while there was no significant difference between the CNN model and tumor size ($p = 0.105$) and AFP level ($p = 0.531$).

Discussion

In this study, we established a CNN model for MVI prediction in HCC by extracting deep features from IVIM-DWI. The results showed that the CNN model achieved good diagnostic performance, suggesting that this CNN model with IVIM-DWI may provide a noninvasive and effective method for MVI prediction. Moreover, our study was based on IVIM sequences, which did not require contrast agents but its diagnostic performance was comparable with previous studies based on contrast-enhanced MRI [19, 20]. Therefore, it can be used in patients with contraindications to contrast agents or renal failure, which is of great clinical significance [21, 22].

Table 2 Performance of clinical characteristics and ADC assessed by ROC analysis in the validation cohort while using the training cohort to determine the cutoff value

Characteristics	Cutoff	Accuracy	Sensitive	Specificity	AUC (95% CI)
AFP level (ng/ml)	1.5	0.700	0.867	0.600	0.733 (0.574–0.892)
Tumor size (cm)	2.25	0.575	0.733	0.480	0.607 (0.426–0.787)
ADC ($10^{-3}\text{mm}^2/\text{s}$)	1.449	0.538	1.000	0.280	0.640 (0.468–0.812)

Table 3 Performance of deep learning models for MVI prediction in the validation cohort

Framework	Accuracy	Sensitive	Specificity	AUC (95% CI)
9 <i>b</i> -value images	0.763 (0.725, 0.800)	0.709 (0.667, 0.733)	0.866 (0.800, 0.880)	0.810 (0.760, 0.829)
IVIM parameter maps	0.600 (0.550, 0.625)	0.560 (0.533, 0.600)	0.650 (0.559, 0.720)	0.590 (0.555, 0.643)
Fusion model	0.775 (0.700, 0.800)	0.666 (0.600, 0.860)	0.870 (0.760, 0.920)	0.829 (0.776, 0.848)

In this study, we evaluated the performance of parameter values for MVI prediction. Previous studies reported that the standard ADC and *D* values were significantly lower in HCCs with MVI by univariate analysis while only the *D* value was an independent risk factor for MVI of HCC by multivariate analysis [11, 23]. However, in our study, only the standard ADC was statistically significant for MVI prediction in the validation cohort. This result might be explained by the fact that the *D* value was obtained by fitting 9 *b*-values IVIM-DWI, which might be severely affected by image artifacts and misalignment between corresponding pixels in multi-*b*-value images [24], while ADC was generated by only fitting *b*₀ and *b*₁₀₀₀. In addition, the tumor size of HCCs in our study was relatively small, so the measurement of the parameter values influenced by artifacts might have been more obvious. This result also reflected that IVIM parameters, obviously affected by motion and artifacts, would have significantly reduced the performance of IVIM for tumor characterization.

A previously published study showed that deep features extracted from DWI through deep convolutional networks were useful for MVI prediction, especially when establishing a fusion model in which deep features were derived from three *b*-value (*b* = 0, *b* = 100, and *b* = 600) images and an ADC map yielded better diagnostic performance with an AUC of 0.79 [15]. This result suggested that deep features extracted from more *b*-value images could better predict MVI. The result of our study that was based on 9 *b*-values IVIM-DWI with an AUC of 0.810 (range 0.760, 0.829) was consistent with that of a previous study and had a better performance than traditional DWI with fewer *b*-values. In addition, the deep learning model we proposed superimposed 9 *b*-value images in the channel dimension, which allowed us to obtain additional characteristics between *b*-values.

Furthermore, we also conducted MVI prediction based on the IVIM parameter maps using the CNN model. In the present study, deep features directly obtained from 9 *b*-value images showed better performance than those from the parameter maps, inferring that image features from 9 *b*-value images can be more representative than those from the parameter maps. This might be attributed to the fact that the IVIM parameter map calculation is based on pixel-by-pixel fitting of the biexponential

function, which is often prone to errors and calculation-intensive, preventing real-time analysis from being performed [10]. To improve predictive performance, the fusion model combined with IVIM-DWI and significant clinical features was also carried out. Our study suggested that the fusion model slightly improved the predictive efficacy for MVI compared with CNN based on IVIM alone, which suggested the potential value of clinical features for MVI prediction.

To create the visual explanation for the CNN model, Grad-CAM was performed. The results reflected that the network fixated most of its attention on the tumor marginal region, which might be related to previous studies showing that nonsmoothed tumor margin was an independent predictor for MVI [25, 26].

To the best of our knowledge, the current routine clinical DWI scans in hospitals are based on 3 *b*-value images, which are mainly used for the detection of HCC lesions, and the fitting generation of ADC parameter maps was used for the characterization of lesions. As a new strategy, the current IVIM-DWI typically needs more *b*-values for accurate fitting of the IVIM-DWI model to obtain *D*, *D*^{*}, and *f* parameter maps for the characterization of lesions. The acquisition of more *b*-value images definitely requires more scanning time. In contrast, this study showed that based on the current multi-*b*-value images of IVIM-DWI, mining potential information through the deep network can obtain better MVI prediction performance than using the 3 *b*-value images method proposed in a previous study [15]. Moreover, this process directly makes predictions based on IVIM-DWI images, without relying on the parameter fitting of traditional IVIM-DWI models. If the purpose of the clinical routine is to detect lesions, the 3 *b*-value images of routine DWI are sufficient. However, for lesion characterization, it is worthwhile to collect more *b*-value images to improve tumor characterization performance.

There were several limitations to this study. First, because the amount of clinical data was small and the slice thickness of DWI scanning was large, we only extracted slices in the axial view of tumors in DWI as the original samples and augmented the data through image resampling. We also realized that there were large overlapping areas in the image patches, which brought the risk of

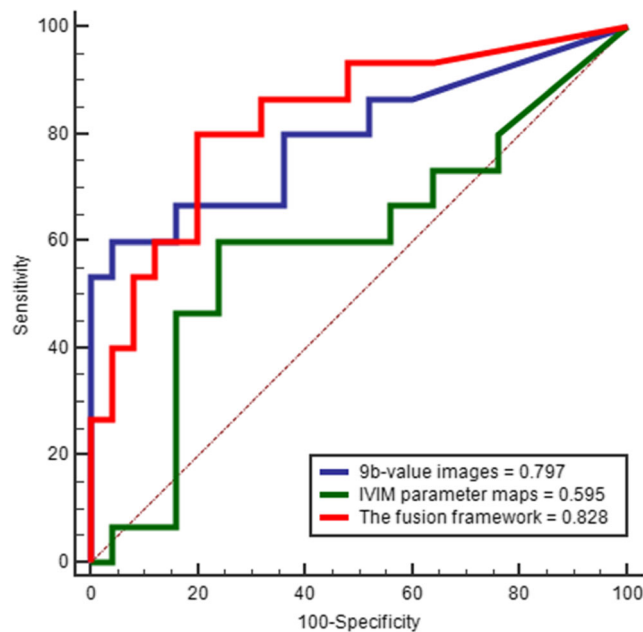


Fig. 4 ROC curves of the deep learning models including 9 b -value images, parameter maps, and the proposed fusion framework for MVI prediction

overfitting. Therefore, we used dropout and early stopping techniques to mitigate the risk of overfitting. Second, the acquisition sequence of clinical IVIM fluctuated greatly

and the range of b -values was uncertain. The quality of image control seriously affects the generalization performance and migration ability of the trained deep learning model. Third, this method only studied MVI prediction with IVIM sequences, and there were other conventional scanning sequences in the clinic, such as T1WI and T2WI. The better fusion of multisequence information may achieve better discrimination, which should be better for clinical applications. Finally, our deep learning model only worked out the characteristics of MVI based on images. In addition to MVI information, HCC also has many indicators such as grading, survival, and prognosis. How to use deep learning based on medical images to accurately predict a variety of pathological and prognostic information of HCC should be the focus of CNN-based deep learning system.

In conclusion, our study suggested that a CNN model based on IVIM achieved a high diagnostic performance for MVI prediction in HCC preoperatively. In addition, the fusion model combined with deep features of IVIM, clinical characteristics (AFP level and tumor size), and ADC achieved better predictive performance than the model only based on IVIM. Thus, we believe that the proposed model can be used to predict MVI and assist clinicians in developing precise treatment strategies.

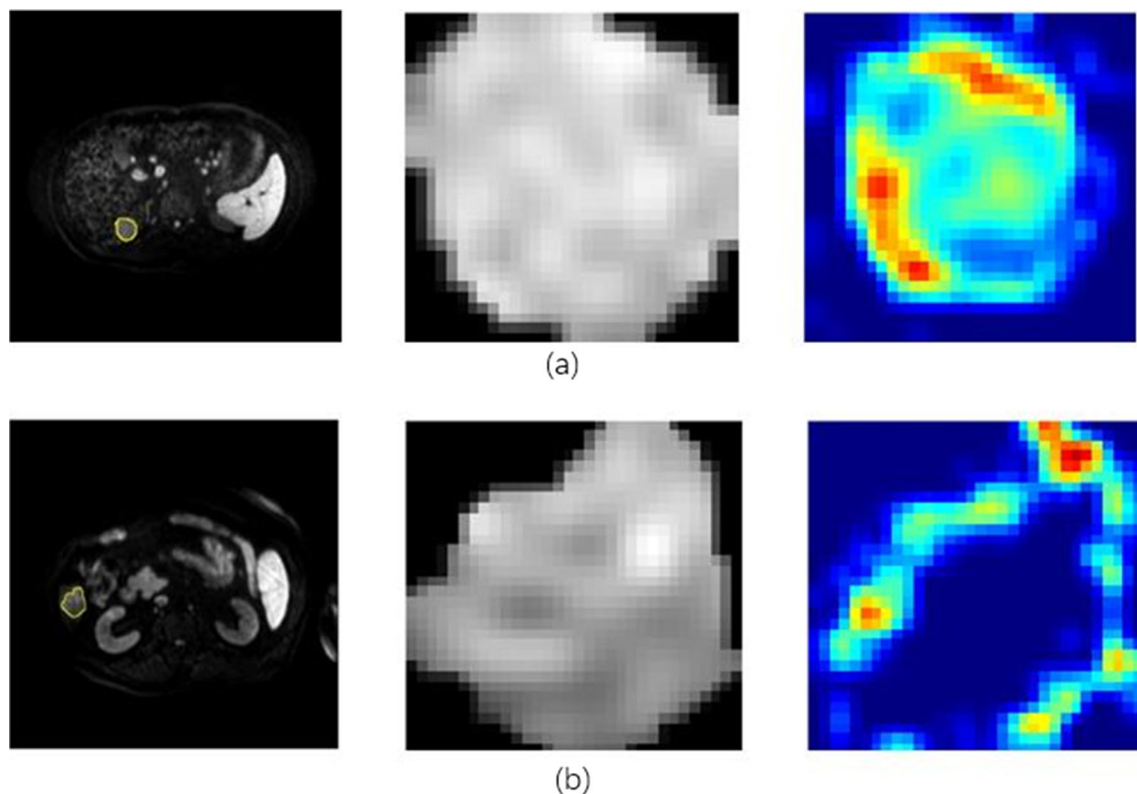


Fig. 5 Attention heatmap drawn by class activation mapping. Darker red indicates greater contribution for MVI prediction, while darker blue indicates lower contribution. **a** A case of a 26-year-old man who was

pathologically confirmed to have HCC without MVI; **b** a case of a 52-year-old man who was pathologically confirmed to have HCC with MVI

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Declarations

Guarantor The scientific guarantor of this publication is Yikai Xu.

Conflict of interest One of the authors (Yuting Liao) is an employee of GE Healthcare and one of the authors (Kan Deng) is an employee of Philips Healthcare. The remaining authors declare no relationships with any companies whose products or services may be related to the subject matter of the article.

Statistics and biometry No complex statistical methods were necessary for this paper.

Informed consent Written informed consent was waived by the Institutional Review Board.

Ethical approval Institutional Review Board approval was obtained.

Study subjects or cohorts overlap In fact, 98 patients in our study overlapped with a previous study (An attention-based deep learning model for predicting Microvascular invasion of hepatocellular carcinoma using an intravoxel incoherent motion model of diffusion-weighted magnetic Resonance imaging). This previous study was a methodological study aimed at proposing a new approach for processing IVIM data and exploring an attentional technique to improve the performance of convolutional neural networks in tumor characterization. However, our current study mainly evaluated the performance of CNN based on IVIM in predicting MVI of HCC lesions. Furthermore, we had included the latest cases to expand the sample size. In addition, our study also evaluated the performance of clinical characteristics, IVIM parameter values, deep learning model based on IVIM parameter maps and the fusion model combined deep features of IVIM, clinical characteristics, and ADC, which could better show the usefulness of the CNN model based on IVIM data for MVI prediction in clinical practice.

Methodology

- retrospective
- diagnostic or prognostic study
- performed at one institution

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